Aman Jaiswal

Founding Engineer

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PROFILE

Founding Engineer at Blocvue, a live multi-tenant compliance platform for UK higher-risk residential buildings. I build AI into domains where a wrong answer has real consequences, and enforce correctness at the data layer rather than in application code. MSc Artificial Intelligence (Distinction, QMUL); designer and sole developer of BillSync, a finance app whose assistant runs entirely on-device.

PROFESSIONAL EXPERIENCE

Founding Engineer — Aug 2026 – Present
Blocvue — London, UK

- Full-stack delivery and platform operations across Next.js, NestJS and PostgreSQL on AWS, for a compliance platform where a missed statutory deadline can be a criminal offence rather than a support ticket. Live since September 2026.
- Multi-tenant model keeps row-level security as the enforcement boundary rather than application code: 100+ tables, every client's data isolated at the database. Designed to scale to 3,000 buildings.
- Built an applicability engine mapping 55 statutory duties across four legal frameworks onto individual buildings, and a versioned document layer holding the golden thread, a statutory record in which nothing can be silently replaced.
- Built the platform's regulatory-change monitor: an AI layer that watches legislation.gov.uk and regulator guidance for changes to statutory obligations, maps each against the obligation register and reports what needs to move. Every flag cited to source, nothing applied without human sign-off.

Founder & Solo Developer — Apr 2026 – Present
BillSync — London, UK

- Designed, built and shipped a personal finance tracker and social bill-splitter to Android and iOS solo: product, backend, mobile client and store releases.
- On-device assistant runs Google's Gemma 4 (E2B, LiteRT-LM, ~2.6 GB) entirely on the phone. It logs expenses, splits bills and settles balances, confirming every action before it runs. No financial question leaves the device and there is no server-side AI.
- Money is computed in the database in exact decimal, never floating point on the client. Shares round to the smallest currency unit with the remainder to the payer, so a bill's shares always sum to the bill.
- Flutter (Dart 3), Riverpod, go_router. Supabase Postgres with row-level security on every table and SQL functions guarding every write to the shared ledger. Deno Edge Functions, Firebase Cloud Messaging, Google Play Billing.

PROJECTS

Anamnesis – AI Patient Intake & Clinical Memory System

- Multi-stage conversational intake using LLaMA-3.3-70B (via Groq), extracting chief complaint, symptom attributes, medications, allergies and safety red flags into validated JSONB in PostgreSQL.
- RAG pipeline over sentence-transformer embeddings and pgvector answers clinician-style questions about patient history, with source attribution on every response.
- FastAPI backend; React and TypeScript frontend separating patient intake from clinician review.

Calma – Offline Multi-Modal AI Emotional Support System

- Fully offline, so nothing a user says leaves their machine. Real-time text and voice.
- Fine-tuned RoBERTa for multi-label emotion classification using a custom Focal Loss trainer in PyTorch and layer-wise learning rate decay; test F1-macro 0.56.
- LLaMA 3.2 for response generation with Whisper (ASR) and F5-TTS (TTS) in a multi-threaded architecture for low-latency turn-taking.

EDUCATION

Queen Mary University of London — Sep 2024 – Sep 2025
MSc in Artificial Intelligence — Distinction
Dissertation: *A Comparative Study of Transformer Architectures for Legal Clause Extraction*. Fine-tuned FLAN-T5 reached 75.91% F1 and 73.44% EM on the CUAD contract benchmark, ahead of fine-tuned LEGAL-BERT (70.74% F1) and zero-shot Gemini 2.5 Pro (67.98% F1).

CHRIST (Deemed to be University) Jun 2021 – Apr 2024
Bachelor of Computer Applications
Event Head, INTERFACE 2024, a national-level undergraduate IT festival.

SKILLS

Languages: Python, TypeScript, JavaScript, Dart, SQL
Backend & Data: PostgreSQL (row-level security, JSONB, pgvector), NestJS, FastAPI, Supabase, Node.js, REST APIs
Frontend & Mobile: Next.js, React, Flutter, Riverpod, Tailwind CSS
AI/ML: LLMs, RAG, fine-tuning, embeddings, vector search, on-device inference (LiteRT-LM), PyTorch, Hugging Face Transformers
Infrastructure: AWS, Docker, Git, Linux